Gymnastiar Al Khoarizmy

Data Science Undergraduate Student
Institut Teknologi Sumatera

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GitHub Profile
LinkedIn Profile

Summary

Final-year Data Science student with hands-on experience in Python, machine learning, deep learning, MLOps, and data validation. Experienced in building applied AI projects involving data preprocessing, annotation workflows, model evaluation, and quality analysis across computer vision, speech understanding, and big data analytics. Interested in contributing to AI data quality and production-oriented data science projects.

Education

- **Bachelor of Science in Data Science** — *2022 – 2026 (Expected)*
  Institut Teknologi Sumatera, Lampung — GPA: 3.64/4.00

Experience

- **Investment Data Analyst Intern** — *Jun 2025 – Sep 2025*
  Ministry of Investment and Downstream Industry / BKPM — Jakarta, Indonesia
  - Developed an Excel VBA-based monitoring tool to integrate and automate quarterly investment reporting compliance checks across company and project records.
  - Built a predictive machine learning model to estimate the transition time of investment projects from construction to production phase.
  - Processed investment datasets, performed feature engineering and model evaluation, and translated analytical results into insights for proactive project monitoring.
  - Presented analytical findings and model results to directors and mentors to support data-driven decision-making.
- **Data Science & Programming Practicum Assistant** — *Jul 2024 – Dec 2025*
  Institut Teknologi Sumatera — Lampung, Indonesia
  - Mentored students across programming, statistics, data mining, and data science-related practicum sessions, including Python, R, regression, classification, hypothesis testing, ANOVA, and model evaluation.
  - Assisted students in debugging code, preprocessing data, implementing statistical and machine learning models, and interpreting analytical results.
  - Evaluated weekly assignments and supported practical learning activities to improve students' technical understanding and problem-solving skills.

Projects

- **Zero-Shot Coastal Waste Detection and Material Classification** — *Final Project / Thesis*
  Grounding DINO, CLIP, Computer Vision, Model Evaluation
  - Built a zero-shot AI pipeline to detect and classify coastal waste materials using Grounding DINO and CLIP.
  - Designed prompt scenarios and evaluated model quality using IoU, precision, recall, F1-score, and classification accuracy.
  - Analyzed detection and classification errors to identify data quality issues, false positives, and model limitations.
- **End-to-End Spoken Language Understanding for Retail Transaction Recording** — *Course Project*
  CNN-BiLSTM, MLOps, Deep Learning — GitHub
  - Built an Indonesian spoken language understanding pipeline for retail transaction recording using deep learning-based sequence modeling.
  - Implemented data preprocessing, model training, evaluation, and documentation in an end-to-end project workflow.
  - Applied MLOps concepts to support reproducibility, structured experimentation, and collaborative model development.
- **Plant Disease Diagnosis System using MLOps and Active Learning** — *INNOVEST 2026 – 3rd Winner*
  MLOps, Active Learning, Data Labeling Workflow
  - Developed a machine learning-based plant disease diagnosis concept supported by MLOps and active learning.
  - Designed an iterative workflow where uncertain predictions can be reviewed, labeled, and reused to improve future model performance.
- **Predictive Flood Analytics with Hadoop Ecosystem in Lampung** — *Course Project*
  Hadoop Ecosystem, Big Data Analytics, Predictive Modeling — GitHub
  - Built a predictive analytics workflow using the Hadoop ecosystem for flood-related data analysis in Lampung.
  - Processed and analyzed multi-source data to support disaster risk analytics and predictive modeling.

Publication

- **Seasonal Forecasting of Ferry Passenger Demand** – IJECS — *Article*

Technical Skills

Programming: Python, R, SQL
Data Science: Data Cleaning, Data Validation, Data Labeling, Feature Engineering, Machine Learning, Deep Learning, Model Evaluation
Libraries: Pandas, NumPy, Scikit-learn, PyTorch, Matplotlib
Tools: Git, GitHub, Docker, Streamlit, Hadoop, Excel VBA, Tableau, Looker Studio
Soft Skills: Attention to Detail, Analytical Thinking, Documentation, Communication, Collaboration
Languages: Indonesian (Native), English

Certifications

- **Python Project for Data Science** – IBM Coursera
- **Python for Data Science, AI & Development** – IBM Coursera